Use-Case Flow Specification

Traffic Management System

Use Case: UC-01 — Monitor Traffic and Plan Route

Primary Actor: Normal User / Driver

Supporting Actors: Mapping/GIS Service, Database, Notification Service

Preconditions

1. The user is registered and authenticated.

2. Traffic and road information is available.

3. The mapping service or map database is available.

4. The database is available to retrieve traffic and route information.

Postconditions

5. The system displays current traffic conditions and congestion levels.

6. A suitable or least-congested route is provided to the user.

7. Relevant traffic alerts are displayed to the user.

Main Success Scenario

1. User logs into the Traffic Management System.

2. System authenticates the user and displays the dashboard.

3. User selects the live traffic map or route-planning option.

4. System retrieves current or simulated traffic and congestion information.

5. System displays roads, traffic density, congestion, and reported incidents on the map.

6. User enters the source and destination.

7. System analyzes available routes using distance, estimated travel time, and congestion information.

8. System identifies a suitable or least-congested route.

9. System displays the recommended route and estimated travel information.

10. System displays relevant traffic alerts, accidents, road closures, or hazards.

11. User views the recommended route and traffic information.

Alternate Flow

4a. Traffic data unavailable: The system displays an appropriate message and uses available/simulated data where applicable.

6a. Invalid source or destination: The system displays a validation error and asks the user to enter valid locations. The use case returns to Step 6.

7a. Mapping/routing service unavailable: The system displays an error message and asks the user to try again later.